Customer Insights – A Statistical Investigation

Project Title:
Unlocking Customer Insights: A Statistical Investigation

Activities:
Apply descriptive statistics, exploratory data analysis, visualizations, and hypothesis testing to investigate behavioural patterns, spending trends, and demographic influences in a real-world customer dataset.

Business Problem Statement:
A mid-sized Indian retail company is sitting on a treasure trove of customer data — but lacks clarity on what story it tells. They need a team of analysts to conduct a rigorous statistical investigation to validate business assumptions, uncover hidden patterns, and support data-driven decision-making.

As a data analytics intern, you have been tasked with:

- Using statistical tools to explore and summarize the dataset
- Comparing customer behavior across demographic segments
- Testing business hypotheses using statistical methods

Key Goals:

- Analyze customer demographics and behavior using descriptive statistics
- Visualize spending, frequency, and demographic attributes
- Segment customers using appropriate quantitative logic
- Test statistical hypotheses around:
- Age and spending
- Gender and transaction frequency
- Geography and engagement
- Deliver clear, statistically valid insights

Dataset Description
You are provided with a synthetic dataset of **10675 rows** for roughly **1,000 unique customers**. The dataset exhibits Behavioural Signals like Spending habits, interaction recency, and lifestyle choices like pet ownership

The dataset description:

Column Name	**Description**
CustomerID	Unique identifier for each customer
Name	Full name of the customer (US names used)

Column Name	Description
State	US state where the customer resides
Education	Highest level of education attained (High School to PhD)
Gender	Gender identity (Male, Female, Non-Binary)
Age	Age of the customer (between 18 and 80)
Married	Marital status: Yes or No
NumPets	Number of pets owned by the customer (0 to 4)
MonthlySpend	Amount spent monthly in USD, generated using a skewed Gamma distribution
DaysSinceLastInteraction	Number of days since last interaction with the company

Step-by-Step Investigation Plan

Note: Wherever Statistical Computation is applicable, you will need to use Python and associated libraries

Step 1: Understand Your Data
Business Purpose: Familiarize yourself with who your customers are and what attributes are available to you.

- Load and preview the dataset.
- Check data types, unique values, and presence of nulls.
- Understand which variables are categorical and which are numerical.

Step 2: Descriptive Statistics
Business Purpose: Describe your customer base — how old are they, how much do they spend, are they active?

- Compute:
  - Mean, median, std dev for Age, MonthlySpend, DaysSinceLastInteraction
  - Mode for categorical variables: Gender, Education, Married

Step 3: Data Visualization
Business Purpose: Reveal patterns that numbers alone can't show.

- Plot histograms and boxplots for Age, MonthlySpend
- Create a bar chart for Gender, Education, State
- Scatterplot: Age vs MonthlySpend
- KDE: Spending behavior by education level or marital status

Step 4: Bivariate Analysis
Business Purpose: Check how customer attributes relate to one another.

- Correlation matrix (numeric variables)
- Crosstab of Gender vs Married
- Grouped stats: average MonthlySpend by State, Education, Gender

Step 5: Formulate Hypotheses
Business Purpose: Turn business questions into statistical tests.

Business Question	**Statistical Test**
Do males and females spend differently?	Independent t-test
Does education level impact average monthly spend?	One-way ANOVA
Is marital status related to the number of pets owned? [Only for DS Students]	Chi-square test
Are older people less active?	Correlation (Age vs DaysSinceLastInteraction)
Does state-wise spend vary significantly?	ANOVA

Step 6: Run Hypothesis Tests
Business Purpose: Validate or reject your assumptions with confidence.

- Define null and alternate hypotheses
- Choose test based on data types
- Check assumptions: normality, independence, homogeneity of variance
- Interpret p-values and confidence intervals

Step 7: Present Business Insights
Business Purpose: Translate stats into strategy.
Create 4–5 takeaways. For example:

- "Customers with Master's degrees spend 18% more per month on average."
- "Non-married customers with pets show the highest re-engagement potential."
- "Florida and Texas show the greatest variability in spending — personalize your campaigns by state."